INCH-POUND
MIL-DTL-24643/56G
27 November 2023
SUPERSEDING
MIL-DTL-24643/56F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 600 VOLTS, TYPE LS2SWL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Watertight)

First	-	Copper conductor, tin-coated 16 AWG, class B stranding, ASTM B8.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation with nominal wall thickness 0.019 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation with one black and one white conductor for each pair.
Fourth	-	Two conductors (one black and one white) cabled together with a nominal lay of 3 inches to form a pair.
Fifth	-	An optional binder.
Sixth	-	Braided shield of 34 or 36 AWG tin-coated copper. Braid angle of 30 to 35 degrees. Minimum coverage of 85 percent.
Seventh	-	Shield insulation of two polyester tapes, outer tape sealed. An alternate shield insulation may consist of one polyester tape plus an extruded jacket of a suitable transparent material; minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2.
Eighth	-	Seven shielded pairs cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers may be used as necessary to make a firm, well-rounded assembly.
Ninth	-	An optional binder.
Tenth	-	Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.075 inch. Overall cable diameter shall be 0.860 inch minimum, 0.910 inch maximum.

NOTE 1: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

NOTE 2: LS2SWL-7 military part number M24643/56-01UN.

Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:17Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	4.66
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2,000
Conductor to shield	1,000
Component shield to component shield	500
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	500
Conductor to shield	500
Component shield to component shield	100
Conductor continuity	No failure
Shield continuity	No failure

Group A:

Visual and dimensional	No failure
Capacitance	
Mutual capacitance (nominal at 1 kilohertz [kHz]) (picofarad [pF]/100 feet)	35
Capacitance unbalance (percent, max.)	10
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C])	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:17Z
Check the source to verify that this is the current version before use.

Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Cable fill (sealant) removability	No failure
Shield (conformance to material, construction, and coverage)	No failure

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Cold working (-4±3.6 °F [-20±2 °C])	No damage
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content (percent, max.)	
Jacket	0.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:17Z
Check the source to verify that this is the current version before use.

Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS2SWL-7	1,500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:17Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

Preparing activity:
Navy – SH
(Project 6145-2013-072)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:17Z
Check the source to verify that this is the current version before use.